CT127-3-2 Programming for Data Analysis

CLO1	Explain the data analytic concepts and techniques with scripting approach (C2, PLO1)	Exam
CLO2	Apply appropriate analytic technique and methods to analyze data in a given domain (C3, PLO2)	Assignment-Analysis
CLO3	Present findings for decision making based on the analysis output (A2,PLO5)	Assignment-Findings

Assignment-Analysis(35%)

Question No.	Topic	Question Vs Taxonomy						PLO
		Cognitive Level						
		1	2	3	4	5	6	
		SQ	SQ	SQ	SQ	SQ	SQ	
Q1	Report Content - Techniques and Methods			35%				2
	Total			35%				

Assignment-Findings(15%)

Question No.	Topic	Question Vs Taxonomy						PLO
		Affective Level						
		1	2	3	4	5		
		SQ	SQ	SQ	SQ	SQ		
Q1	Report Finding and Discussion			15%				5
	Total			15%				

			Fail	Marginal Fail	Pass	Credit	Distinction
ANALYSIS (70%) [CLO2-PLO2]	**ANALYSIS TECHNIQUES (Methodology)**		**0 - 3**	**4**	**5-6**	**7**	**8-10**
			• Not done • Complete out of the scope • Irrelevant techniques proposed.	• Inappropriate techniques proposed. • Just mentioned the technique without any description.	• Appropriate techniques proposed with brief description	• Appropriate techniques proposed with description • Minimal justification provided.	• Good techniques proposed with detailed description • Excellent justification provided
			0-11	**12-14**	**15-19**	**20-22**	**23-30**
	ANALYSIS METHODS	**Complexity**	• Not done • Incomplete work. • Very basic RScript provided without any analysis.	• Minimal analysis done. • Some mixture of exploration and visualization process. • Some data pre-processing requires but not done. • Poor programming practices	• A good mixture of exploration and visualization process. • Incorporate many basic concepts (descriptive analysis only) • Analysis and visualization done by executing existing functions with original data. • Basic statistical analysis carried out • Basic programming practices shown.	• Good mixture of exploration, manipulation and visualization process. • Incorporate many intermediate concept / knowledge/analysis methods (from descriptive to predictive analysis) • Good programming practices shown.	• A good mixture of exploration, manipulation and visualization process implemented • Incorporate many advanced concepts / knowledge/analysis methods (from descriptive to prescriptive analysis). • Full coverage without any omission or limitations. • Excellent programming practices and analysis done.
		Supporting Document	**0 - 7**	**8 - 9**	**10 - 12**	**13 – 14**	**15 – 20**
			• Not available • Irrelevant explanation / supporting documents provided	• Just mentioned the analysis method or concepts applied. • Minimal explanation / supporting documents provided	• Brief explanation of the concepts, analysis method applied • Some screenshots attached with brief explanation.	• Good explanation of the concepts, analysis method applied • All screenshots attached with explanation.	• Excellent explanation of the concepts, analysis method applied • All screenshots attached with detailed explanation. • No weakness identified.
			0 - 3	**4**	**5-6**	**7**	**8-10**
	ADDITIONAL FEATURES		• No additional features • Additional features added to the existing analysis are meaningless.	• Additional features do not enhance the result of the analysis.	• One additional feature included to enhance the result of the analysis	• Two additional features included to enhance the result of the analysis	• Three additional features included to enhance the result of the analysis. • Additional features implemented ease the analysis process.
							/70

		Fail	Marginal Fail	Pass	Credit	Distinction
FINDING AND DISCUSSION (30%) [CLO2-PLO5]		**0 - 7**	**8 - 9**	**10 - 12**	**13 - 14**	**15 - 20**
	Finding and Discussion	• Not done. • Describe the results only. • Irrelevant discussion	• Describe the results only. • Poor / minimal discussion / interpretation of the result obtained.	• Finding were discussed briefly without any recommendation.	• Good discussion on the finding. • Objectives identified were reviewed. • Some recommendation provided	• Detailed discussion on the finding based on the objectives identified. • Provided a detailed view on future direction • Effective interpretation done with professional quality recommendation
		0 - 3	**4**	**5-6**	**7**	**8-10**
	Project Introduction and report structure and standard	• Documentation not done. • Mostly poor documentation standards with most of the contents / components are missing. • Very poor command of the language, etc. • Irrelevant domain and scenario.	• Some missing components • Poor documentation standards and works are messy. • Poor command of the language.	• Overall average documentation standards. • No missing sections / components. • Brief content. • Average command of the language and flow of information from one section to another section.	• Good documentation standards. • Good explanation and description shown in the content. • Good flow of information from one section to another section.	• Very good documentation standards. • Detailed explanation and description shown in the content. • Clear information and precise content with good information flow. • Overall documentation has a professional outlook.
						/30

Generative AI usage Declaration

Criterion	Requirement	Fully Met (100%)	Mostly Fully Met (80- 99%)	Several Partial (60-79%)	Multiple Not Met (40-59%)	Major failures / dishonesty (0-39%)	Penalty
AI tools clearly identified and AI declaration submitted	Declaration included where required Specific tools named (e.g., ChatGPT, Copilot, Gemini).	No penalty	1-3%	4-6%	7-8%	9-10%	
AI usage adequately explained	Student explains where/how AI was used	No penalty	1-5%	6-10%	11-15%	16-20%	
AI Prompt and output	Records of AI-assisted work processes / AI interaction records	No penalty	1-7%	8-16%	17-23%	24%-30%	
AI output verified/refined	AI-generated content properly checked, corrected, edited, validated and integrated	No penalty	1-5%	6-10%	11-15%	16-20%	
Evidence of human contribution	Student contributed thinking, analysis, synthesis, restructuring, interpretation, etc.	No penalty	1-5%	6-10%	11-15%	16-20%	
Fabricated citations / data	No hallucinated references or false content	No penalty	Disciplinary referral				
						Disciplinary referral if the penalty exceed 70% **Total**	

Noted:

- Fabrication includes unverifiable, AI-generated, or incorrect citations, sources, or data presented as factual.
- Contribution levels are estimated based on proportion of original student work.
- Markers will assign penalties at the upper, middle, or lower end of the range based on the extent and clarity of evidence provided.
- Penalties across criteria are not strictly additive. The final penalty reflects the overall severity and pattern of issues identified.
- Borderline cases will be resolved using professional academic judgment supported.